



Course Name

Computer Programming Lab

Lab Title

Open Ended Lab

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Task 1: Division Calculator

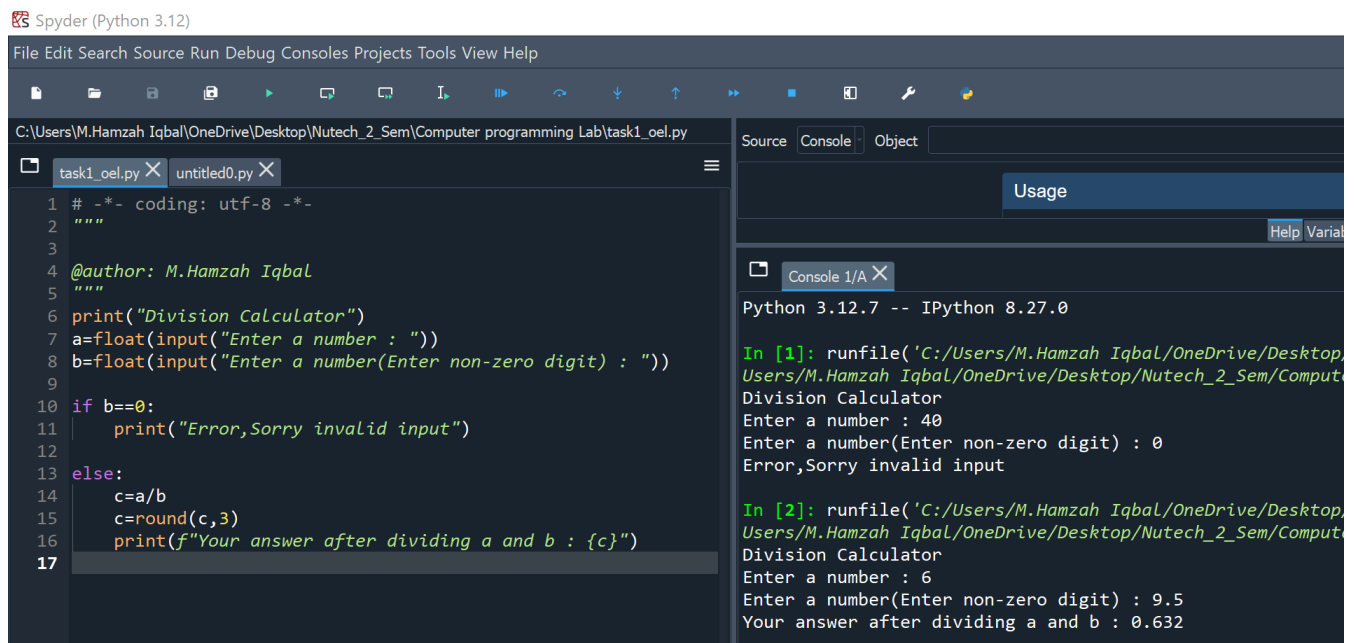
Write a program that asks the user to enter two numbers and calculates their division. Handle the ZeroDivisionError exception if the user enters 0 as the denominator. Display an appropriate error message and ask the user to enter a non-zero denominator.

Code:

```
print("Division Calculator")
a=float(input("Enter a number : "))
b=float(input("Enter a number(Enter non-zero digit) : "))

if b==0:
    print("Error,Sorry invalid input")

else:
    c=a/b
    c=round(c,3)
    print(f"Your answer after dividing a and b : {c}")
```



The screenshot shows the Spyder Python IDE interface. The left pane displays the code for the division calculator. The right pane shows the console output, which includes the program's execution and the user's input for two different test cases.

```
1 # -*- coding: utf-8 -*-
2 """
3
4 @author: M.Hamzah Iqbal
5 """
6 print("Division Calculator")
7 a=float(input("Enter a number : "))
8 b=float(input("Enter a number(Enter non-zero digit) : "))
9
10 if b==0:
11     print("Error,Sorry invalid input")
12
13 else:
14     c=a/b
15     c=round(c,3)
16     print(f"Your answer after dividing a and b : {c}")
17
```

Console 1/A X

Python 3.12.7 -- IPython 8.27.0

In [1]: runfile('C:/Users/M.Hamzah Iqbal/OneDrive/Desktop/Users/M.Hamzah Iqbal/OneDrive/Desktop/Nutech_2_Sem/Comput...')
Division Calculator
Enter a number : 40
Enter a number(Enter non-zero digit) : 0
Error,Sorry invalid input

In [2]: runfile('C:/Users/M.Hamzah Iqbal/OneDrive/Desktop/Users/M.Hamzah Iqbal/OneDrive/Desktop/Nutech_2_Sem/Comput...')
Division Calculator
Enter a number : 6
Enter a number(Enter non-zero digit) : 9.5
Your answer after dividing a and b : 0.632

Figure 1 Input and Output

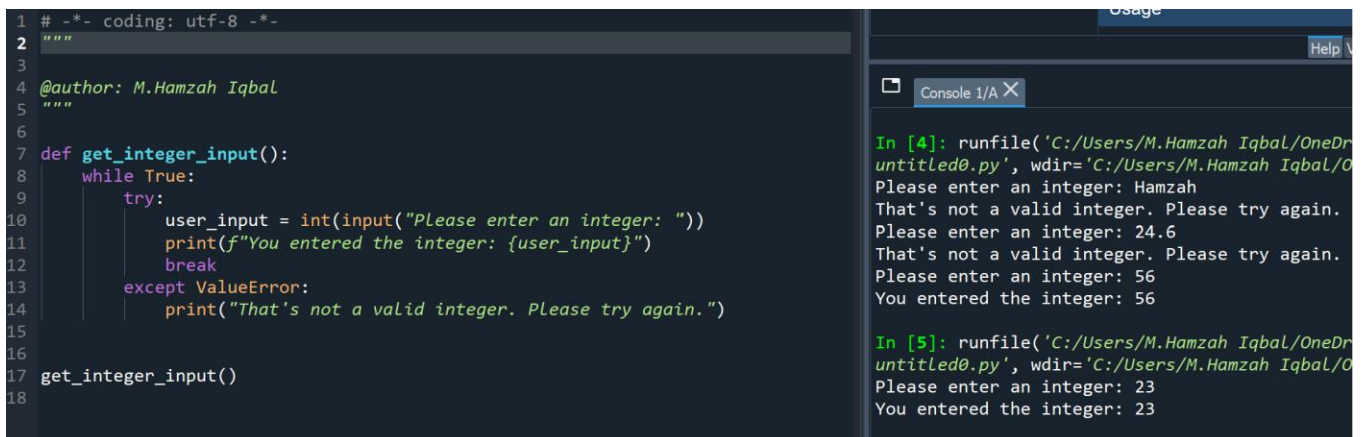
Task 2: Integer Input

Write a program that asks the user to enter an integer.
Handle the ValueError exception if the user enters something other than an integer.
Display an appropriate error message and ask the user to enter a valid integer.

Code:

```
def get_integer_input():
    while True:
        try:
            user_input = int(input("Please enter an integer: "))
            print(f"You entered the integer: {user_input}")
            break
        except ValueError:
            print("That's not a valid integer. Please try again.")

get_integer_input()
```



The screenshot displays a code editor on the left and a console window on the right. The code editor shows the Python function `get_integer_input()` defined with a `while True` loop and a `try/except` block to handle `ValueError` exceptions. The console window shows the output of the function when it is run, demonstrating the program's behavior for both valid and invalid integer inputs.

```
1 # -*- coding: utf-8 -*-
2 """
3
4 @author: M.Hamzah Iqbal
5 """
6
7 def get_integer_input():
8     while True:
9         try:
10             user_input = int(input("Please enter an integer: "))
11             print(f"You entered the integer: {user_input}")
12             break
13         except ValueError:
14             print("That's not a valid integer. Please try again.")
15
16
17 get_integer_input()
18
```

Console 1/A X

In [4]: runfile('C:/Users/M.Hamzah Iqbal/OneDr
untitled0.py', wdir='C:/Users/M.Hamzah Iqbal/O
Please enter an integer: Hamzah
That's not a valid integer. Please try again.
Please enter an integer: 24.6
That's not a valid integer. Please try again.
Please enter an integer: 56
You entered the integer: 56

In [5]: runfile('C:/Users/M.Hamzah Iqbal/OneDr
untitled0.py', wdir='C:/Users/M.Hamzah Iqbal/O
Please enter an integer: 23
You entered the integer: 23

Figure 2 Input and Output

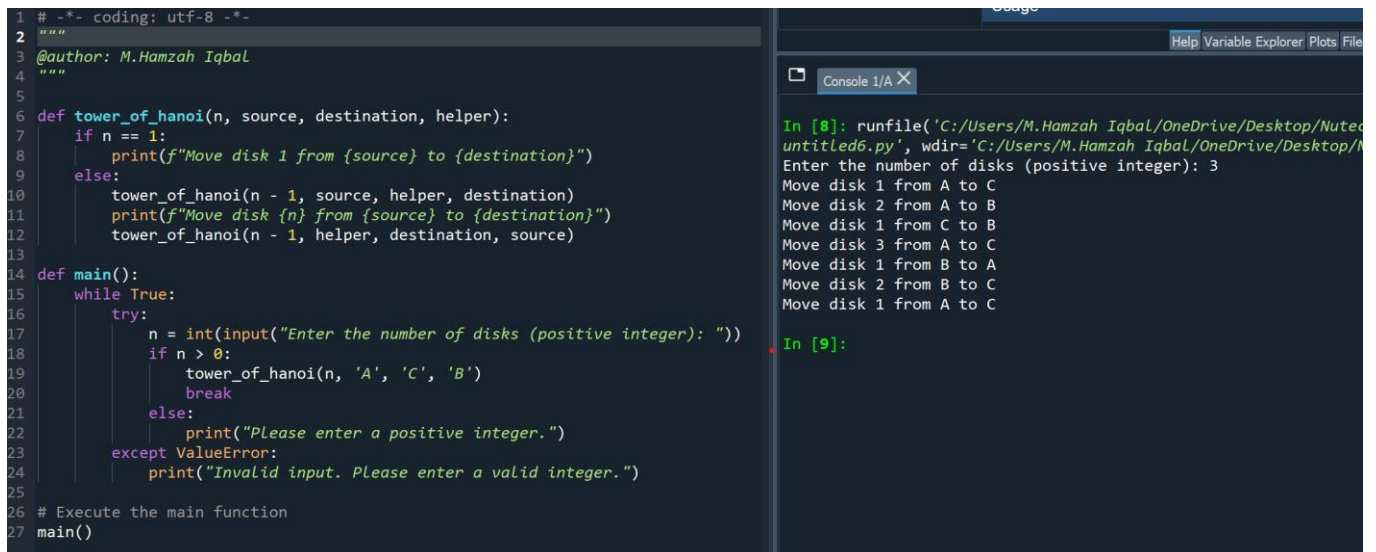
Towers of Hanoi:

Code:

```
def tower_of_hanoi(n, source, destination, helper):
    if n == 1:
        print(f"Move disk 1 from {source} to {destination}")
    else:
        tower_of_hanoi(n - 1, source, helper, destination)
        print(f"Move disk {n} from {source} to {destination}")
        tower_of_hanoi(n - 1, helper, destination, source)

def main():
    while True:
        try:
            n = int(input("Enter the number of disks (positive integer): "))
            if n > 0:
                tower_of_hanoi(n, 'A', 'C', 'B')
                break
            else:
                print("Please enter a positive integer.")
        except ValueError:
            print("Invalid input. Please enter a valid integer.")

# Execute the main function
main()
```



The screenshot displays a Jupyter Notebook interface with two main panels. The left panel shows the Python code for the Towers of Hanoi problem, including a recursive function `tower_of_hanoi` and a `main` function that handles user input and error handling. The right panel shows the console output of the code. The user has entered '3' for the number of disks, and the program has executed the recursive function, resulting in the following sequence of moves: 'Move disk 1 from A to C', 'Move disk 2 from A to B', 'Move disk 1 from C to B', 'Move disk 3 from A to C', 'Move disk 1 from B to A', 'Move disk 2 from B to C', and 'Move disk 1 from A to C'.

```
1 # -*- coding: utf-8 -*-
2 """
3 @author: M.Hamzah Iqbal
4 """
5
6 def tower_of_hanoi(n, source, destination, helper):
7     if n == 1:
8         print(f"Move disk 1 from {source} to {destination}")
9     else:
10         tower_of_hanoi(n - 1, source, helper, destination)
11         print(f"Move disk {n} from {source} to {destination}")
12         tower_of_hanoi(n - 1, helper, destination, source)
13
14 def main():
15     while True:
16         try:
17             n = int(input("Enter the number of disks (positive integer): "))
18             if n > 0:
19                 tower_of_hanoi(n, 'A', 'C', 'B')
20                 break
21             else:
22                 print("Please enter a positive integer.")
23         except ValueError:
24             print("Invalid input. Please enter a valid integer.")
25
26 # Execute the main function
27 main()
```

Console 1/A X

In [8]: runfile('C:/Users/M.Hamzah Iqbal/OneDrive/Desktop/Nuted6/untitled6.py', wdir='C:/Users/M.Hamzah Iqbal/OneDrive/Desktop/Nuted6')
Enter the number of disks (positive integer): 3
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
In [9]:

Figure 3 Input and Output